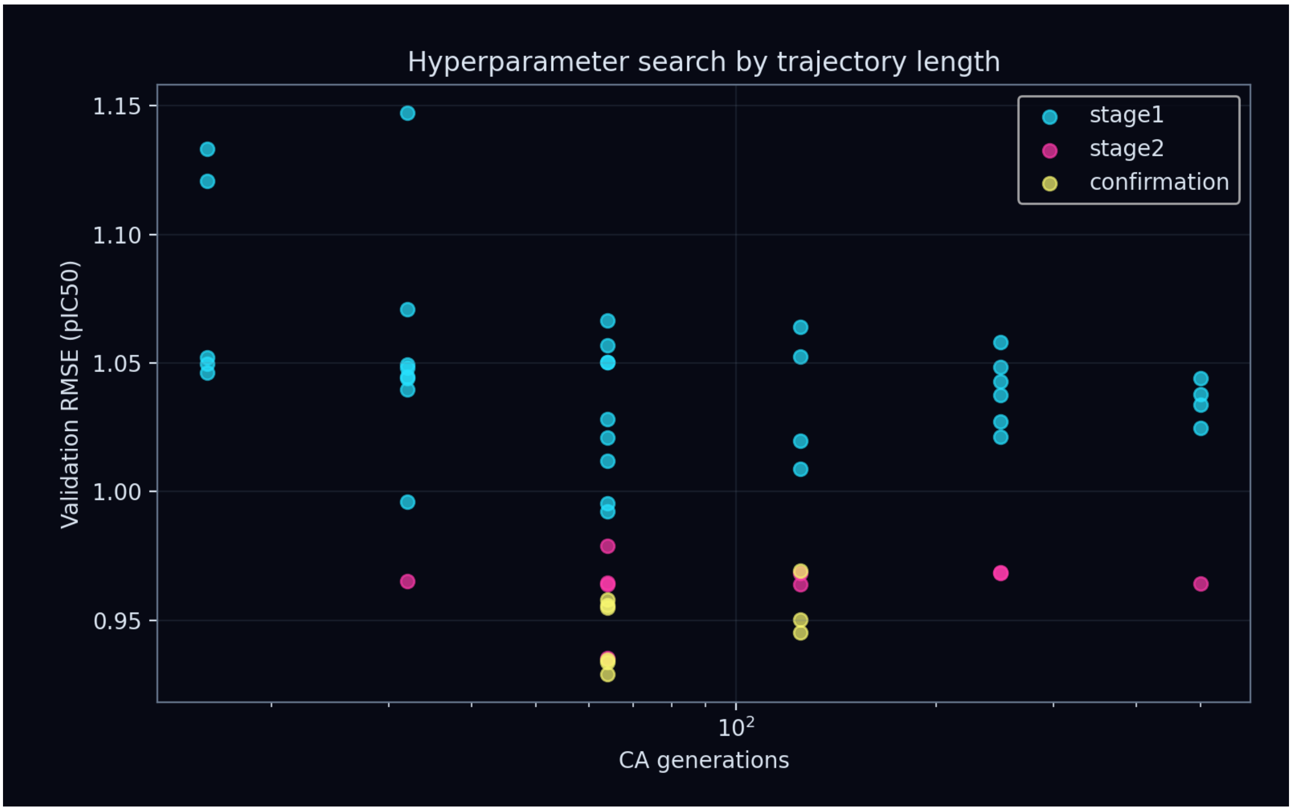
Strange Matter Engine

Production study: conservative_graph_flux

Direct-inhibition pIC50 | grouped validation | blinded set excluded

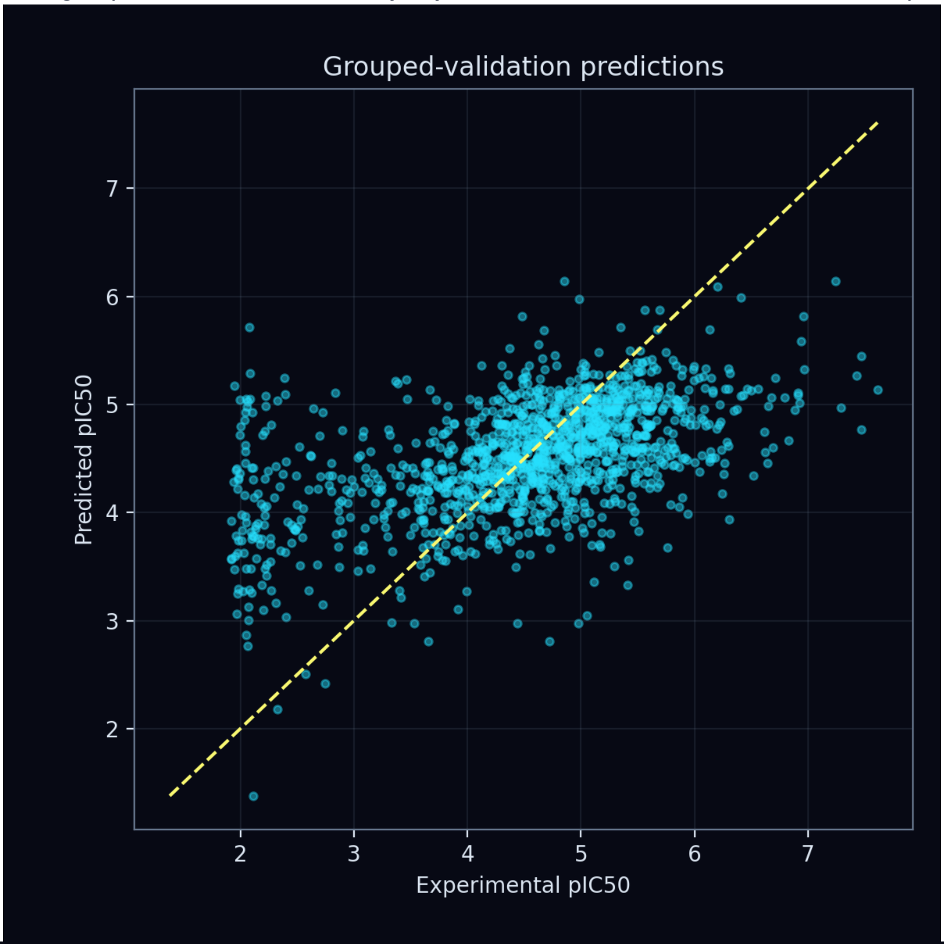
Quantity	Result
Validation RMSE	0.9251 pIC50
Fit RMSE	0.8991 pIC50
Generations	64
Dynamical channels	8
CA learning rate	0.003
Ridge penalty	0.01
CA L2	0.0001
Gradient clip	1.0
Confirmation mean RMSE	0.9324
Confirmation seed SD	0.0025

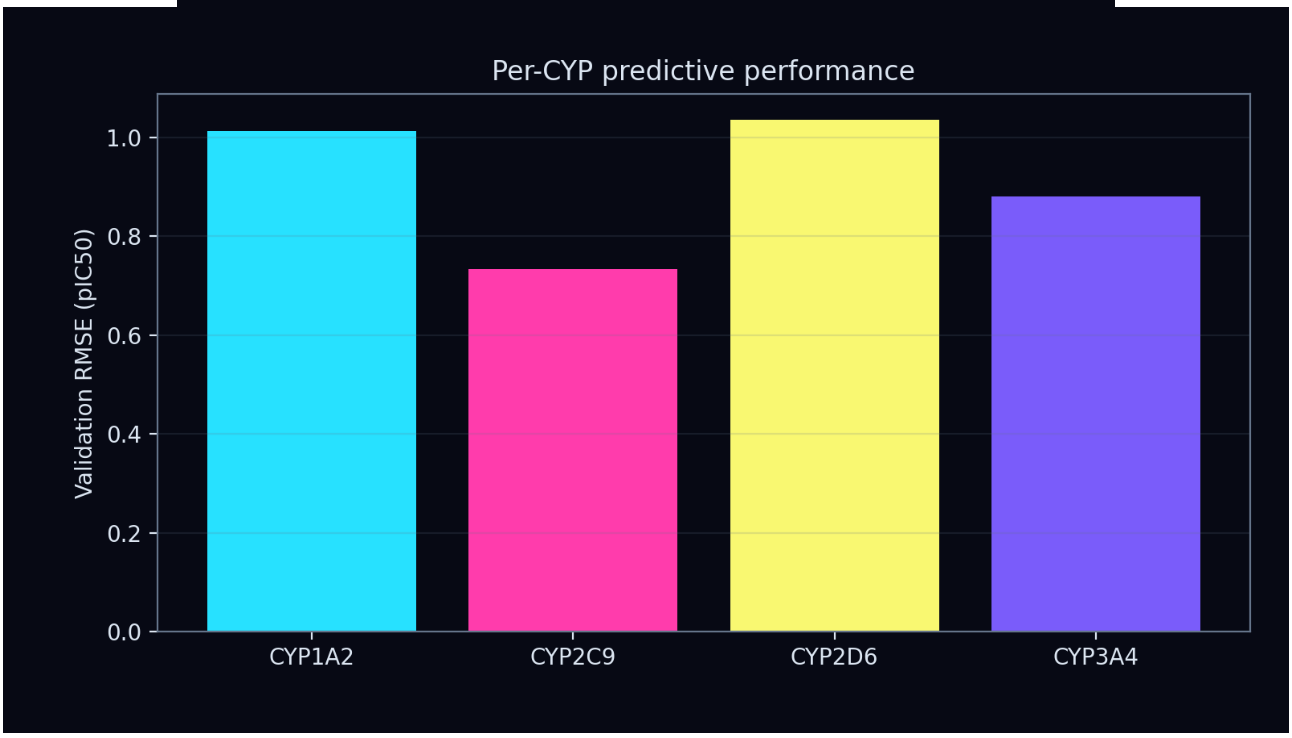
Enhanced CA controls: update scale 0.4, initial-state scale 1.5, support fraction 0.75, bond temperature 0.5. Rule dynamics: A=0.2, B=0.8, C=0.5, D=0.05.



Predictive validation

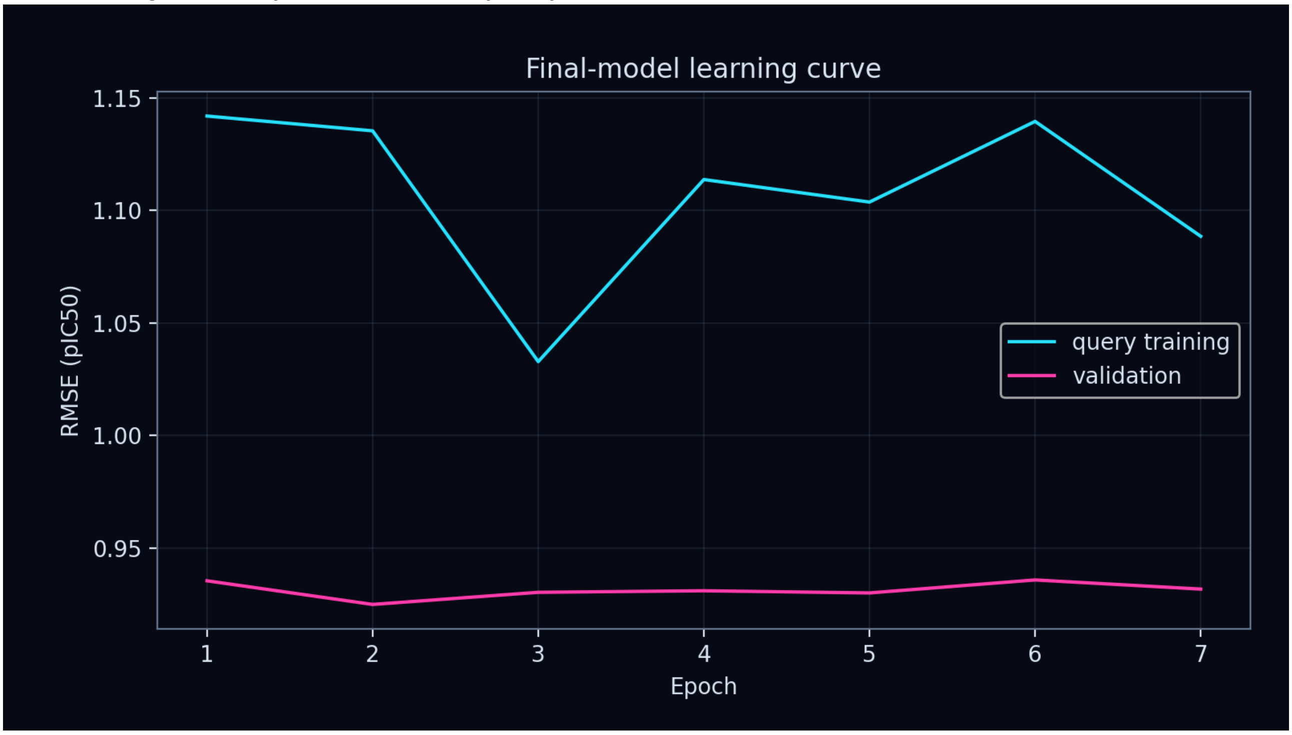
Model selection used grouped-validation RMSE only. Dynamical-interest scores did not influence promotion.





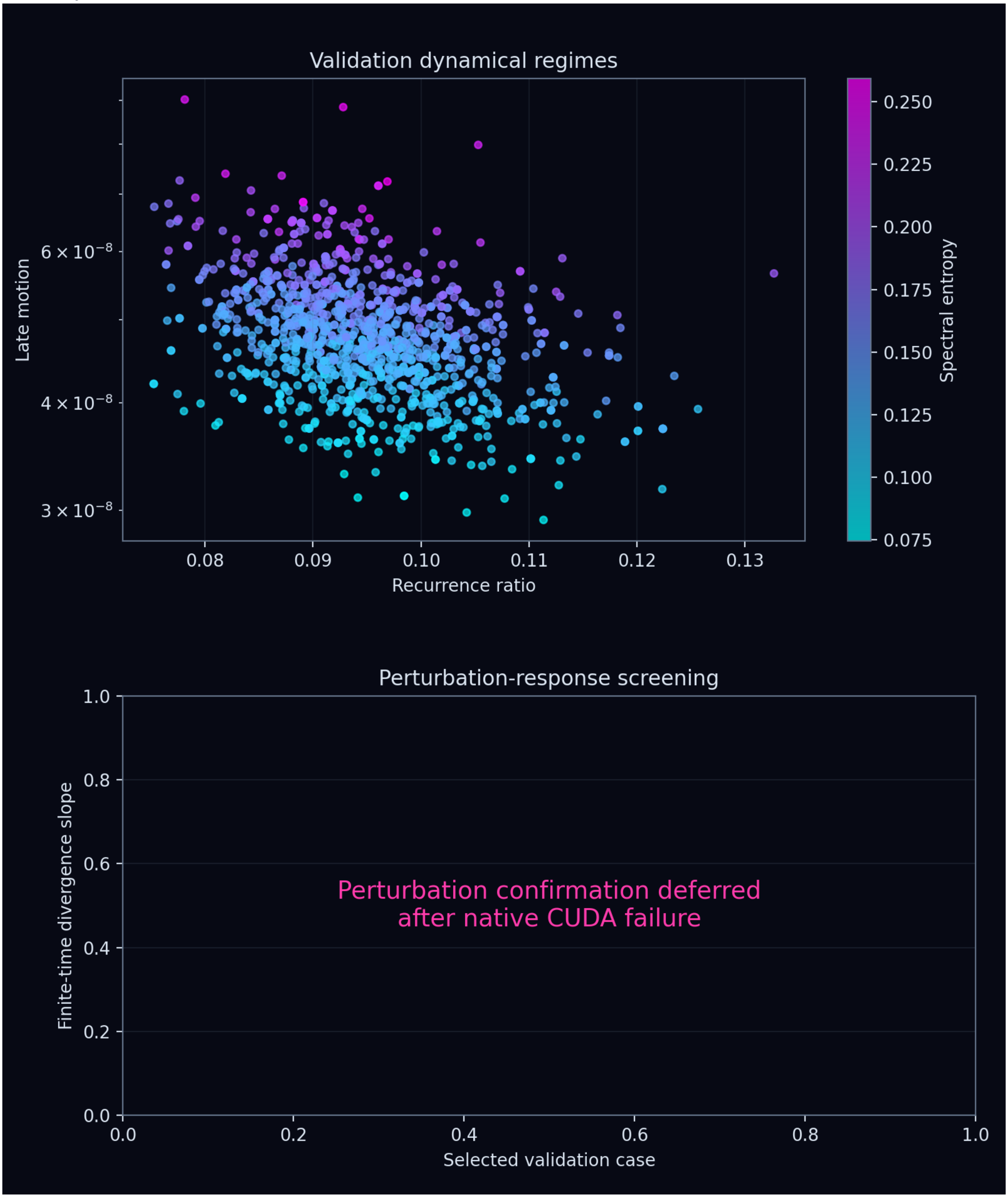
Optimization

Ridge coefficients and the unpenalized intercept were solved from permitted fitting fingerprints. Query error differentiated through the solve into the graph CA; Adam updated only CA parameters. Bond-conditioned messages, cosine learning-rate decay, and multitime trajectory statistics were shared across all five rules.



Emergent dynamics

The validation trajectories were screened for convergence, recurrence, persistent motion, spectral complexity, and perturbation sensitivity. Labels ending in candidate require longer, renormalized and seed-repeated confirmation before any attractor or chaos claim.



Scientific limitations

This report describes one transition-rule study under the declared grouped split. Hyperparameter selection and early stopping used labelled training data only. The blinded challenge set was not loaded or predicted. Finite-time positive slopes are screening signals rather than proof of a strange attractor or sustained chaos.